

Module 5

Clustering Algorithms

Clustering:-

- Cluster analysis is unsupervised learning.

Def: A Technique of partitioning data into meaningful group

Def:

A Technique of partitioning collection of unlabeled dataset that have many attributes into meaningful disjoint clusters.

- Clustering is done using trial and error process.

Differences between clustering & classification

clustering

- Unsupervised learning and cluster formation are done by trial and error as there is no supervisor
- Unlabelled data
- No prior knowledge in clustering
- Cluster results are dynamic

classification

- Supervised learning with the presence to provide training and testing data.
- labelled data
- Knowledge of the domain is must to label the samples of dataset.
- once ~~the~~ a label is assigned, it does not change.

Applications:-

- Grouping based on customer buying pattern
- Profiling of customers based on lifestyle.
- In information retrieval applications.
- Identifying the groups of genes that influence a disease

Categorical Variable:

• It is just the code or symbol to represent the values

$$\text{Distance}(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

Ordinal Variables:

• Ordinal variables are just like categorical values but with an inherent order.

$$\text{Distance}(x, y) = \frac{|\text{Position}(x) - \text{Position}(y)|}{n-1}$$

Cosine Similarity:-

$$\sin(x, y) = \frac{x \cdot y}{|x| |y|} = \frac{\sum_{i=1}^n x_i \cdot y_i}{\sqrt{\sum_{i=1}^n x_i^2} \cdot \sqrt{\sum_{i=1}^n y_i^2}}$$

eg: $A = \{1, 1, 0\}$ what is the cosine similarity.
 $B = \{0, 1, 1\}$

$$A \cdot B = 1 \times 0 + 1 \times 1 + 0 \times 1 = 1$$

$$A = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$$

$$B = \sqrt{0^2 + 1^2 + 1^2} = \sqrt{2}$$

$$\sin(A, B) = \frac{1}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} = 0.5$$

Hierarchical clustering Algorithms.

① Place each N sample or data instance into a separate cluster. So, initially N clusters are available.

② Repeat the following steps until a single cluster is formed.

(a) Determine two most similar clusters.

(b) Merge the two clusters into a single cluster reducing the number of clusters as $N-1$.

③ Choose resultant clusters of step 2 as result.

Q. Consider the array of points as shown below

objects	X	Y
0	1	4
1	2	8
2	5	10
3	12	18
4	14	28

Apply single linkage algorithm

Sol.

Proximity Matrix

objects	0	1	2	3	4
0	-	4.1231	7.211	17.805	27.29
1		-	3.605	14.142	15.620
2			-	10.63	20.124
3				-	10.198
4					-

$$\text{distance}(0,1) = \sqrt{(1-2)^2 + (4-8)^2} = \sqrt{1^2 + 4^2} = \sqrt{17} = 4.1231$$

$$\text{distance}(0,2) = \sqrt{(1-5)^2 + (4-10)^2} = \sqrt{4^2 + 6^2} = 2\sqrt{13} = 7.2111$$

$$\text{distance}(0,3) = \dots$$

Module-5

Clustering Algorithms

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cosine similarity:-

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eg:- $A = \{1, 1, 0\}$ & $B = \{0, 1, 1\}$ what is the cosine similarity.

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$$A = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$$

$$B = \sqrt{0^2 + 1^2 + 1^2} = \sqrt{2}$$

$$\sin(A, B) = \frac{1}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} = 0.5$$

Hierarchical Clustering Algorithms:-

- ① Place each N samples or data instance into a separate cluster, so, initially, N clusters are available.
- ② Repeat the following steps until a single cluster is formed
 - (a) Determine two most similar clusters.
 - (b) Merge the two clusters into a single cluster reducing the number of clusters as $n-1$
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Q:- Consider the array of points as shown below. Apply single linkage algorithm.

objects	x	y
0	1	4
1	2	8
2	5	10
3	12	18
4	14	28

Proximity matrix.

	0	1	2	3	4
0	—	4.1231	7.211	17.80	27.29
1		—	3.605	14.142	15.620
2			—	10.63	20.124
3				—	10.198
4					—

Hence, the object 1 & 2 are clustered together.

Further the process will be ~~for~~ for remaining objects.

	(1,2)	0	3	4
(1,2)	-	4.123	10.630	20.124
0		-	17.804	27.295
3			-	10.198
4				-

$$[(1,2), 0] \Rightarrow (1,0) \quad (2,0) \\ \min(4.123, 7.211) = 4.123$$

$$[(1,2), 3] \Rightarrow (1,3) \quad (2,3) \\ \min(14.142, 10.630) = 10.630$$

$$[(1,2), 4] \Rightarrow (1,4) \quad (2,4) \\ \min(23.324, 20.124) = 20.124$$

$$[0, 3] \rightarrow 17.804$$

$$[0, 4] \rightarrow 27.295$$

$$[3, 4] \rightarrow 10.198$$

The cluster found is (0,1,2)

	(0,1,2)	3	4
(0,1,2)	-	10.630	20.124
3		-	10.198
4			-

$$[(0,1,2), 3] \Rightarrow (0,3) \quad (1,3) \quad (2,3)$$

$$\min(17.804, 14.142, 10.630) = 10.630$$

$$[(0,1,2), 4] \Rightarrow (0,4) \quad (1,4) \quad (2,4)$$

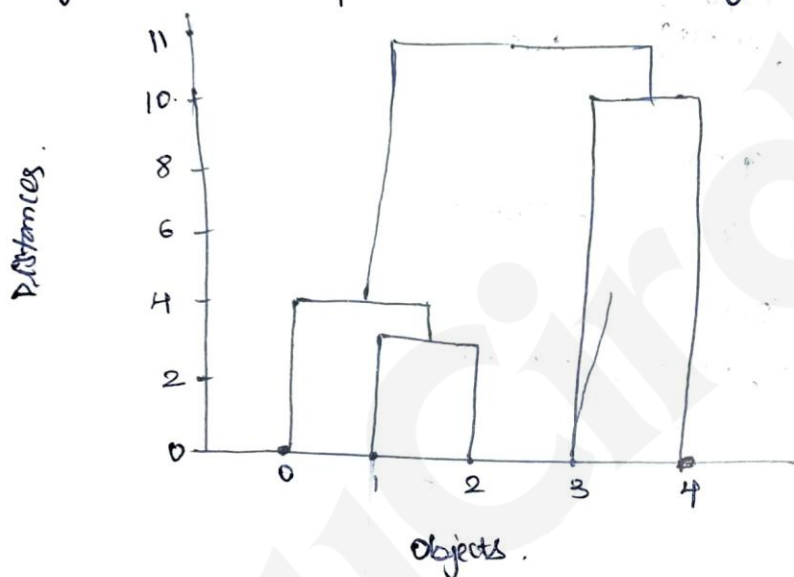
$$\min(27.295, 23.324, 20.124) = 20.124$$

The cluster formed is (3,4)

	(0,1,2)	(3,4)
(0,1,2)	-	
(3,4)		-

↑
 Not required because all ~~clusters~~ objects are clustered hence
 this is the stopping point of cluster.

Dendrogram: pictorial representation of clusters generated.



(V2)
 Complete Linkage Algorithm

- In complete linkage algorithm the distance between 2 objects are computed with maximum of distances.

$$D_c(C_i, C_j) = \max_{a \in C_i, b \in C_j} d(a, b)$$

(V3)
 Average Linkage Algorithm.

- The average distance of all pairs of points across the clusters is used to form clusters.

$$D_{AL}(C_i, C_j) = \frac{1}{m_i m_j} \sum_{a \in C_i, b \in C_j} d(a, b)$$

Mean-shift Algorithm.

- ① Design a window.
- ② Place the window ~~over~~ ^{on} a set of data points.
- ③ Compute the mean for all the points that come under the window.
- ④ Move the center of the window to the mean computed in step 3. Thus, the window moves towards the dense regions. The movement to the dense region is controlled by a mean shift vector. The mean shift vector is given as:

$$V_s = \frac{1}{K} \sum (x_i - x)$$

Here K is the no. of points,

S_K is the data points whose distance from data points x_i & centroid of the kernel x is within the radius of the sphere. Then, the centroid is updated as $x = x + V_s$.

- ⑤ Repeat the steps 3-4 for convergence. Once convergence is achieved, no further points can be accommodated.

Advantages:

- No model assumptions.
- Suitable for all non-convex shapes.
- Only one parameter of the window, that is, bandwidth, is required.
- Robust to noise.
- No issues of local minima or premature termination.

Disadvantages:

- Selecting Bandwidth is a challenging task. If it
 - larger — clusters are missed.
 - smaller — points are missed, and convergence occurs.
- The number of clusters cannot be specified & user has no control over this parameter.

K-Means Algorithm:-

- ① Determine the number of clusters before the algorithm is started. This is called K .
- ② Choose K instances randomly. These are initial cluster centers. → Table
- ③ Compute the mean of the initial cluster ~~centers~~ and assign the remaining sample to the closest cluster based on Euclidean distance or any other distance measure between the instances & the centroid of the clusters.
- ④ Compute new centroid again considering the newly added samples.
- ⑤ Perform the steps 3, 4 till the algorithm becomes stable with no more changes in assignment of instances & clusters.

~~Grid-based~~

→ solve the problem.

Grid based approaches

3 grid based schemes:

- ① Subspace clustering.
- ② concept of dense cells.
- ③ Monotonicity ~~property~~ property.

① ~~Subspace clustering~~

Dense cell clustering:-

- ① Define a set of grid points and assign the given data points on the grid.
- ② Determine the dense and sparse cells. If the number of points in a cell exceeds the threshold value τ , the cell is categorized as dense cell. Sparse cells are removed from the list.
- ③ Merge the dense cells if they are adjacent.
- ④ ~~Take~~ Form a list of grid cells for every subspace as output.

CLIQUE Algorithm

Stage 1:

- ① Identify the dense cells.
- ② Merge the dense cells c_1 & c_2 if they share the same interval.
- ③ Generate Apriori rule to generate $(k+1)^{th}$ cell for higher dimension. Then, check whether the number of points cross the threshold. This is ~~repeated~~ repeated till there are no dense cells or new generation of dense cells.

Stage 2:

- ① Merging of dense cells into a ~~clust~~ cluster is carried out in each subspace using maximal regions to cover dense cells. The maximal region is an hyperrectangle where all cells fall into.
- ② Maximal region tries to cover all dense cells to form clusters.

Note:

Apriori rule/property: One can conclude k -dimensional cell has σ points (attributes), if and only if every $k-1$ dimensional projections of this cell has atleast σ points.

Reinforcement Learning.

14.01 to 14.10.

Overview:

- RL is a branch of ML inspired by how humans learn from interacting with their environment.
 - Like a child ~~to~~ learning through experiences.
 - RL agents learn by receiving feedback in the form of rewards or punishments to improve future decisions.
- *
*
*
↓
important for RL

Scope of RL:

- Useful for solving problems with more uncertainty like Gold Games.

Characteristics of RL:-

- ① Sequential Decision making: Reaching a goal involves a series of decisions, one wrong move can cause failure.
- ② Delayed Feedback: Rewards are not immediate; success or failure is realized after several steps.
- ③ ~~Inter~~ Interdependent Actions.
- ④ Time Related.

RL challenges:

- ① Reward Design: It's difficult to define appropriate rewards and their values.
- ② Lack of a model: ~~Many~~
- ③ Partial observability
- ④ Time-consuming
- ⑤ Complexity - Games with vast configurations (like Go) lack labeled data.

RL Applications:

- Industrial automation.
- Resource management.
- Traffic light control.
- Deepmind applications.
- Recommendation systems.
- Ad bidding.
- Customized applications.
- Driverless cars.
- Games like chess & Go.

Difference between Reinforcement learning vs Supervised learning.

Reinforcement learning

- No supervisor and labeled dataset initially.
- Decisions are dependent and are made sequentially.
- Feedback is not instantaneous and delayed by time.
- Agent action affects the next input data.
- No target values, only goal oriented.

eg: Chess, Go

Supervised learning

- Presence of supervisor and labeled data.
- Decisions are independent of each other and based on input given in the training phase.
- Usually the feedback is instantaneous once the model is created.
- Dependent of initial input to the input given at start.
- Target class is predefined by the problem.

eg: - classifiers

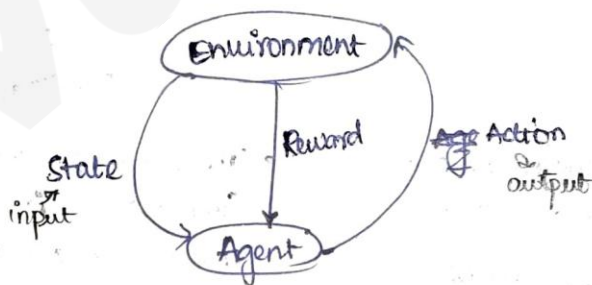
RL

- Mapping from input to output is present.
- Get constant feedback from environment.

Unsupervised learning

- Not present.
- No feedback from environment.

components of Re-inforcement learning :-



Basic components of RL:-

- **Environment**: where actions take place.
- **Agent**: Learner or decision-maker.
- **State**: The current situation of the environment.
- **Action**: What the agent does.
- **Reward**: Feedback from the environment based on the agent's action.

Environment and agent:

- The environment defines inputs, outputs, and rules.
- The agent observes the environment & makes decisions.

States & Actions:-

- The state is the input and action is the output.
- Action lead to transitions between states with associated rewards.

Policies

- A policy is a function that maps the states to actions, guiding the agent's behavior.

2 main types:

- ① Deterministic policy: Always returns the same action for a given state.

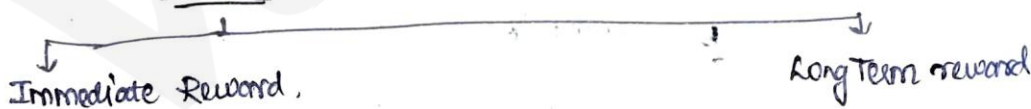
$$a = \pi(s)$$

- ② Stochastic policy: Returns a probability distribution over actions for a given state.

$$\pi(a|s) = \Pr[a=a|s=s]$$

- The goal of the agent is to learn an optimal policy that selects the best actions.
- A better policy yields higher cumulative expected rewards over time, leading to higher state values when optimal.

Rewards:



- After each action, the environment delivers an immediate reward.
- The total reward is the sum of all rewards from the start to the end of an episode.

$$G_t = r_{t+1} + r_{t+2} + r_{t+3} + \dots + r_T$$

G_t is the return

r_T is final reward.

- The total future reward from a move is the sum of the reward at that move plus all rewards from subsequent moves until the episode ends.

- If there's no terminal state we use a discount factor to reduce the impact of future rewards.

discounted return =

$$G_t = \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$$

Markov Decision Process:- (14.5)

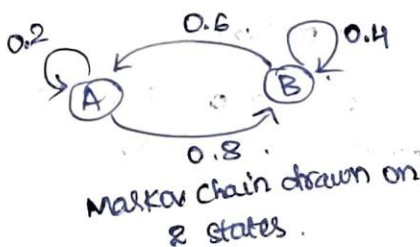
Markov chain?

→ A Markov chain is a stochastic process satisfying the Markov property.
It is a sequence of random variables satisfying conditional probability.

Markov Decision Process.

→ This is an extension towards the Markov chain.

Ex:-



Description:-

Assume 2 universities:-

Eighty percent of students of university A leave for university B for getting a master's degree and only 20% remain in the same university. Similarly 60% move to university A.

Q:-



Assume the universities' hold on a town is initially 60% & 40%. So, what would be the prediction after one month, that is, the hold of university after a year. After two years, will the hold remain same or change? Assume the ~~known~~ rest of the probability transmission information.

Sol:- The starting distribution is U_0 .

$$U_0 = (60\%, 40\%) = (0.6, 0.4)$$

Transition matrix:

$$P = \begin{bmatrix} \text{existing} \downarrow & \text{moved} \downarrow \\ 0.2 & 0.8 \\ 0.4 & 0.6 \end{bmatrix}$$

Prediction/Calculation

According to Markov chain $U_{n+1} = U_n \times P$

Prediction after a year is $U_1 = U_0 \times P$

$$\Rightarrow U_1 = [0.6, 0.4] \begin{bmatrix} 0.2 & 0.8 \\ 0.4 & 0.6 \end{bmatrix}$$

$$= [0.28 \quad 0.72]$$

The hold of universities will be 28% & 72% after 1st year

Second year prediction is $U_2 = U_1 \times P$

$$\Rightarrow [0.28 \quad 0.72] \begin{bmatrix} 0.2 & 0.8 \\ 0.4 & 0.6 \end{bmatrix} = [0.344 \quad 0.656]$$

Hence, hold of universities will be 34.4% & 65.6% in second year

will not remain the same

Markov Decision process depends on

- ① A set of states
- ② A set of actions
- ③ Reward function R
- ④ Policy π
- ⑤ Value V

Multi Arm Bandit Problem:

There are two types of sub problems in Reinforcement problem:

- ① The main problem is the prediction of total reward or return. This is called prediction, policy evaluation or value estimation.

②

$Q_k(a) = \frac{r_1 + r_2 + r_3 + \dots + r_k}{k}$ This function is called action-value function or Q function. This indicates the value of taking a particular action 'a'.

Selection procedure:-

- Exploration
- Exploitation

ϵ -greedy algorithm:-

- ① Generate a random number p in the range $0-1$.
- ② If $p \geq \epsilon$ the algorithm chooses the action with highest reward value.
- ③ If $p < \epsilon$, actions are chosen randomly.
- ④ Update the value of action and repeat for all actions.

With probability ϵ , the action a is chosen randomly, but the rest of the time with probability $1-\epsilon$, the best actions based on greedy approach are selected.

Reinforcement Learning Agent types:-

An agent can be characterized using these methods:

① Value based approaches.

② Policy-based approaches

③ Hybrid methods. $\rightarrow$ value based + Policy based.

④ Model-based methods. } have small methods or algorithms

⑤ Model free methods.

Value based approaches:-

Goal: Optimize the value function $V(s)$.

Def: $V(s)$ is the maximum expected future reward for an agent starting in state s , discounted factor γ

Usage: The agent chooses actions to maximize this value.

Policy based approaches:

Goal: Find the optimal policy π , which is a mapping from states to actions.

Def: The policy gives a probability distribution over actions to each state.

Actor-critic method.

Type $\rightarrow$ Hybrid.

Def: Combines both value based & policy based

Usage: Actor updates the policy, and critic evaluates it using a value function.

Model based approach:

Def: Involves constructing a model of the environment.

Usage: Use the model to predict future states & rewards.

Ex: Markov Decision process (MDP).

Model-free methods:

Definition: No explicit model of the environment is used.

Usage: Uses methods like temporal difference (TD) learning or Monte Carlo simulations to learn from experience.

Purpose: Simulate the model virtually for learning.

Algorithm Selection:-

Model	Environment	On/off	Algorithm -
Present	Static	off	Dynamic programming,
Absent	Dynamic	on	SARSA
Absent	Dynamic	-	Temporal learning, Monte Carlo method
Absent	Dynamic	off	Q-learning.

Temporal Difference learning:

- ① Initialize policy π
 - ② set $V(s) = 0$ for all states $s \in S$
 - ③ Repeat for each episode.
 - ④ Initialize s
 - ⑤ Repeat for each step of episode
 - ⑥ Choose an action 'a' and execute it, Observe the new state and new reward
 - ⑦ Update value using Bellman equation $V(s_t) = V(s_t) + \alpha (G_t - V(s_t))$
 - ⑧ Update as s' .
 - ⑨ Until s is terminal state
- $\alpha \rightarrow$ constant
 $t \rightarrow$ some time
 $s \rightarrow$ state
 $V \rightarrow$ value function
 $G \rightarrow$ returns.

Monte-Carlo Simulation Algorithm:

Assumptions:

- ① All episodes should terminate. No matter from where we start, the episode should end.
- ② MC methods use value-action functions. These values are computed only after the completion of the episode. So, MC methods are incremental methods.

Algorithm:

- ① For a policy π , select state s
- ② Generate an episode
- ③ compute average returns for every first occurrence of state s in an episode.
- ④ update the returns and start a new episode. Repeat for all episodes.

TD.

TD.

MC.

- Exploits markov principle
- Does not require termination

- It is easy to understand & does not exploit markov principle.
- Requires termination & episodes.

Q-learning:

Def: Q-learning is an off policy method because Q is updated based on policies and it better guarantees maximum returns.

To compute total return/reward the following informations are necessary.

- ① Starting state.
- ② Action.
- ③ Reward.
- ④ New state.

• A Q-table can be constructed and filled with random values..

• Q-learning involves 2 policies.

- ① learning policy.
- ② update policy.

• Q-learning is done by methods such as greedy method, softmax or softmax ~~plus~~ plus methods.

• The update mechanism used in Q-learning is as follows.

- ↳ It uses Temporal difference learning and.
- ↳ Bellman equation method.

↓

$$Q_t(s_t, a_t) = Q_t(s_t, a_t) + \alpha TD_t(s_t, a_t)$$

Q-learning algorithm: (Refer textbook).

Problems:

Apply Naive Bayes Algo

Q. Given a test data (Outlook=Sunny, Temp=cool, Hum=high, Windy=False), Apply Naive Bayes to predict the class.

Outlook	Temperature	Humidity	Windy	playTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	mild	High	False	Yes
Rainy	cool	Normal	False	Yes
Rainy	cool	Normal	True	No
Overcast	cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	cool	Normal	False	Yes
Rainy	mild	Normal	False	Yes

Step 1: Calculate prior probability.

PlayTennis $\rightarrow$ Yes - 6
 $\rightarrow$ No - 4

prior probability for PlayTennis=Yes $\rightarrow \frac{6}{10}$

prior probability for PlayTennis=No $\rightarrow \frac{4}{10}$

Step 2: Calculate frequency matrix & likelihood for all features.

Attribute: Outlook.

Outlook	PlayTennis(Yes)	PlayTennis(No)
Sunny	1	3
Overcast	2	0
Rainy	3	1
Total	6	4

Likelihood probability:-

Outlook	P(PlayT=Yes)	P(PT=No)
Sunny	$P(\text{Out=S} \text{PT=Yes}) = \frac{1}{6}$	$P(\text{Out=S} \text{PT=No}) = \frac{3}{4}$
Overcast	$P(\text{Out=O} \text{PT=Yes}) = \frac{2}{6}$	$P(\text{Out=O} \text{PT=No}) = 0$
Rainy	$P(\text{Out=R} \text{PT=Yes}) = \frac{3}{6}$	$P(\text{Out=R} \text{PT=No}) = \frac{1}{4}$

Attribute: Temperature

	PT(Yes)	PT(No)
Temp	1	2
Hot		1
mild	2	
cool	3	1
	<u>6</u>	<u>4</u>

likelihood:

	P(PT=Yes)	P(PT=No)
Temp	1/6	2/4
Hot		1/4
mild	2/6	1/4
cool	3/6	

Attribute: Humidity

	PT(Yes)	PT(No)
Humid	2	3
High	4	1
Normal	<u>6</u>	<u>4</u>

likelihood:

	P(PT=Yes)	P(PT=No)
Humid	2/6	3/4
High		1/4
Normal	4/6	

Attribute: Windy

	PT(Yes)	PT(No)
Windy	1	2
True	5	2
False	<u>6</u>	<u>4</u>

likelihood:

	P(PT=Yes)	P(PT=No)	$P(H/E) = \frac{P(E H) \cdot P(H)}{P(E)}$
Windy	1/6	2/4	
True		2/4	
False	5/6		

Step 3:

$$P(PT=Yes | \text{Test data}) = P(O=Su | PT=Yes) \times P(Tem=cool | PT=Yes) \times P(Hum=high | PT=Yes) \times P(W=T | PT=Yes) \times P(PT=Yes)$$

$$= \frac{1}{6} \times \frac{3}{6} \times \frac{2}{6} \times \frac{1}{6} \times \frac{6}{10} = \frac{1}{360} = 2.7777 \times 10^{-3} = 0.0027777$$

$$\begin{aligned}
 P(PF=No | \text{Test data}) &= P(O=Sun | PT=No) \times P(T=cool | PF=No) \times P(H=high | PT=No) \\
 &\quad \times P(W=T | PT=No) \times P(PT=No) \\
 &= \frac{3}{4} \times \frac{1}{4} \times \frac{3}{4} \times \frac{2}{9} \times \frac{4}{10} \\
 &= \frac{9}{320} = 0.028125
 \end{aligned}$$

Since $P(PT=No | \text{Test data})$ is greater, the test data results in Play Tennis is No.

Q: Apply Naive Bayes

Test data: True, False, False True.

	Toothed	Hair	Breathes	Legs	Species
1	True	True	True	True	mammal
2	True	True	True	True	mammal
3	True	False	True	False	reptile
4	False	True	True	True	mammal
5	True	True	True	True	mammal
6	True	True	True	False	mammal
7	True	False	True	False	Reptile
8	True	False	True	True	Reptile
9	True	True	True	True	mammal
10	False	False	True	True	reptile

Species = M $\rightarrow 6/10$

Species = R $\rightarrow 4/10$

Attribute: Toothed.

Toothed	S=M	S=R
True	5	3
False	$\frac{1}{6}$	$\frac{1}{4}$

Circularood -

Toothed	S=M	S=R
True	5/6 5/6	3/4
False	1/6	1/4

Attribute: Hair

Hair	S=M	S=R	G/G	0
True	6	0	0	4/4
False	0	4	0	
	<u>6</u>	<u>4</u>		

Attribute: Breathes

Breathes	S=M	S=R	G/G	4/4
True	6	4	0	0
False	0	0		
	<u>6</u>	<u>4</u>		

Attribute: Legs

Legs	S=M	S=R	G/G	1/4
True	6	1	0	3/4
False	0	3		
	<u>6</u>	<u>4</u>		

$$P(S=M | \text{test data}) = P(T=\text{true} | S=M) \times P(H=\text{false} | S=M) \times P(B=\text{false} | S=M) \times P(L=\text{True} | S=M) \times P(S=M)$$

$$= \frac{5}{6} \times 0 \times 0 \times \frac{6}{6} \times \frac{6}{10} = 0$$

$$P(S=R | \text{test data}) = P(T=\text{true} | S=R) \times P(H=\text{false} | S=R) \times P(B=\text{false} | S=R) \times P(L=\text{True} | S=R) \times P(S=R)$$

$$= \frac{3}{4} \times \frac{4}{4} \times 0 \times \frac{1}{4} \times \frac{4}{10} = 0$$

• It does not belong to any of the species category.

Q1: Design a perceptron network to implement Norrgate choose w1 & w2 as 0.2 & 0.2 respectively and bias as -0.2. use a learning rate $\alpha = 0.4$

Norrgate:

x1	x2	Yexp	Test	Error	Status	w1	w2
0	0	1	$(0 \times 0.2 + 0 \times 0.2) - (0.2) \Rightarrow 1$	0	NC	0.2	0.2
0	1	1	$0 \times 0.2 + 1 \times 0.2 + 0.2 \Rightarrow 1$	0	NC	0.2	0.2
1	0	1	$1 \times 0.2 + 0 \times 0.2 + 0.2 \Rightarrow 1$	0	NC	0.2	0.2
1	1	0	$1 \times 0.2 + 1 \times 0.2 + 0.2 \Rightarrow 1$	-1	Change	-0.2	

$$\Delta w_1 = 0.4(-1)(1) = -0.4 \Rightarrow 0.2 - 0.4 = -0.2$$

$$\Delta w_2 = 0.4(-1)(1) = -0.4 \Rightarrow 0.2 - 0.4 = -0.2$$

Epoch 2.

α_1	α_2	Yes-	Yes+	Err	State	w_1	w_2
0	0	1	$(0 \times -0.2 + 0 \times -0.2) - (-0.2) \Rightarrow 1$	0	NC	-0.2	-0.2
0	1	1	$(0 \times -0.2 + 1 \times -0.2) - (-0.2) \Rightarrow 0$	0	NC	-0.2	-0.2
1	0	1	$(1 \times -0.2) + (0 \times -0.2) - (-0.2) \Rightarrow 1$	0	NC	-0.2	-0.2
1	1	0	$(1 \times -0.2) + (-0.2) - (-0.2) \Rightarrow 0$	0	NC	-0.2	-0.2